Electronic Circuits and Applications

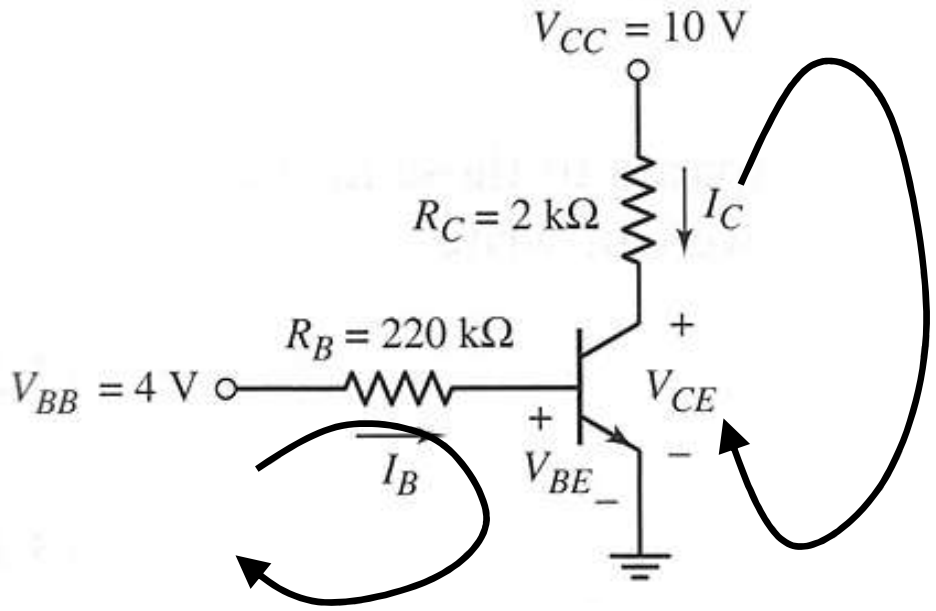
Examples

BJT DC Analysis

Common-Emitter Circuit

Example 1

Calculate the base, collector and emitter currents and the C-E voltage for a common-emitter circuit by considering $V_{BB} = 4$ V, $R_B = 220\text{k}\Omega$, $R_C = 2$ kΩ, $V_{CC} = 10$ V, $V_{BE}(\text{on}) = 0.7$ V and $\beta = 200$.



Solution: The base current is found as

$$I_B = \frac{V_{BB} - V_{BE}(\text{on})}{R_B} = \frac{4 - 0.7}{220} \Rightarrow 15\,\mu\text{A}$$

The collector current is

$$I_C = \beta I_B = (200)(15\,\mu\text{A}) \Rightarrow 3\,\text{mA}$$

and the emitter current is

$$I_E = (1 + \beta) \cdot I_B = (201)(15\,\mu\text{A}) \Rightarrow 3.02\,\text{mA}$$

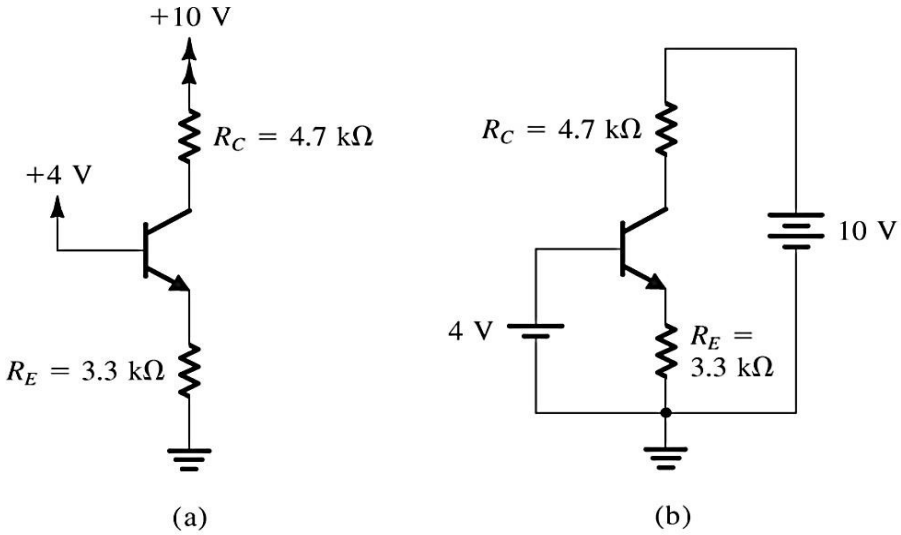
The collector-emitter voltage is

$$V_{CE} = V_{CC} - I_C R_C = 10 - (3)(2) = 4\,\text{V}$$

Example 2

Example 2

$\alpha = 0.99$



KVL at BE loop: $0.7 + I_E R_E - 4 = 0$

$I_E = 3.3 / 3.3 = \underline{\mathbf{1\ mA}}$

Hence, $I_C = \alpha\, I_E = \underline{\mathbf{0.99\ mA}}$

$I_B = I_E - I_C = \underline{\mathbf{0.01\ mA}}$

KVL at CE loop: $I_C R_C + V_{CE} + I_E R_E - 10 = 0$

$V_{CE} = 10 - 3.3 - 4.653 = \underline{\mathbf{2.047\ V}}$

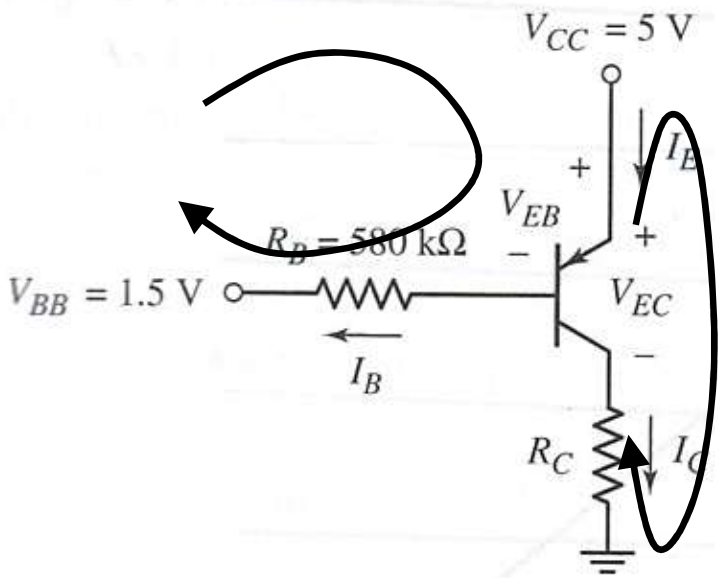
Common-Emitter Circuit - PNP

Example 3

Find I_B, I_C, I_E and R_C such that $V_{EC} = 2.5$ V for a common-emitter circuit .

Consider:

$V_{BB} = 1.5$ V, $R_B = 580$ kΩ, $V_{CC} = 5$ V, V_{EB} (on) $= 0.6$ V, $\beta = 100$.



Solution: Writing a Kirchhoff voltage law equation around the E–B loop, we find the base current to be

$$I_B = \frac{V_{CC} - V_{EB}(\text{on}) - V_{BB}}{R_B} = \frac{5 - 0.6 - 1.5}{580} \Rightarrow 5\,\mu\text{A}$$

The collector current is

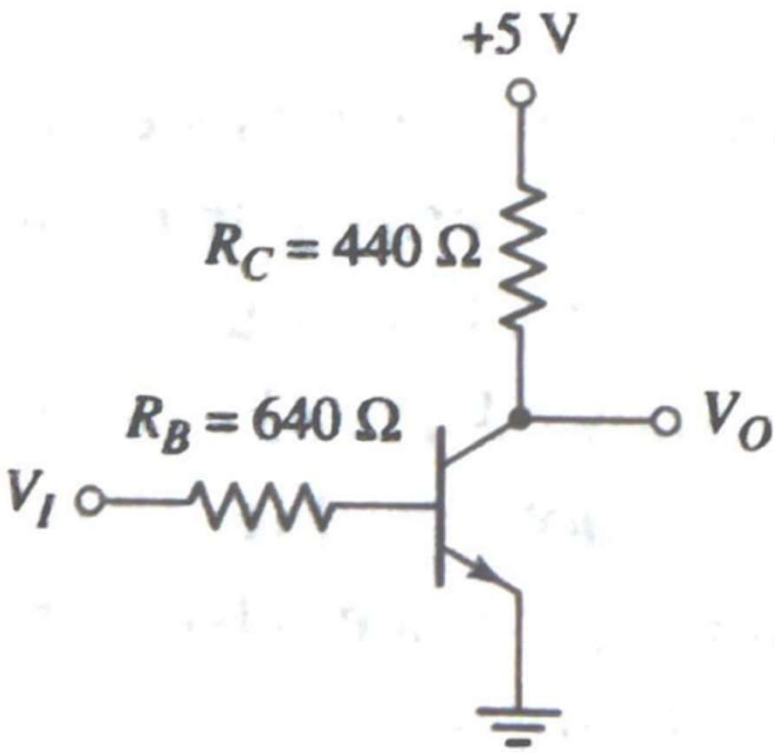
$$I_C = \beta I_B = (100)(5\,\mu\text{A}) \Rightarrow 0.5\,\text{mA}$$

and the emitter current is

$$I_E = (1 + \beta) I_B = (101)(5\,\mu\text{A}) \Rightarrow 0.505\,\text{mA}$$

For a C–E voltage of $V_{EC} = \frac{1}{2} V_{CC} = 2.5$ V, R_C is

$$R_C = \frac{V_{CC} - V_{EC}}{I_C} = \frac{5 - 2.5}{0.5} = 5\,\text{k}\Omega$$



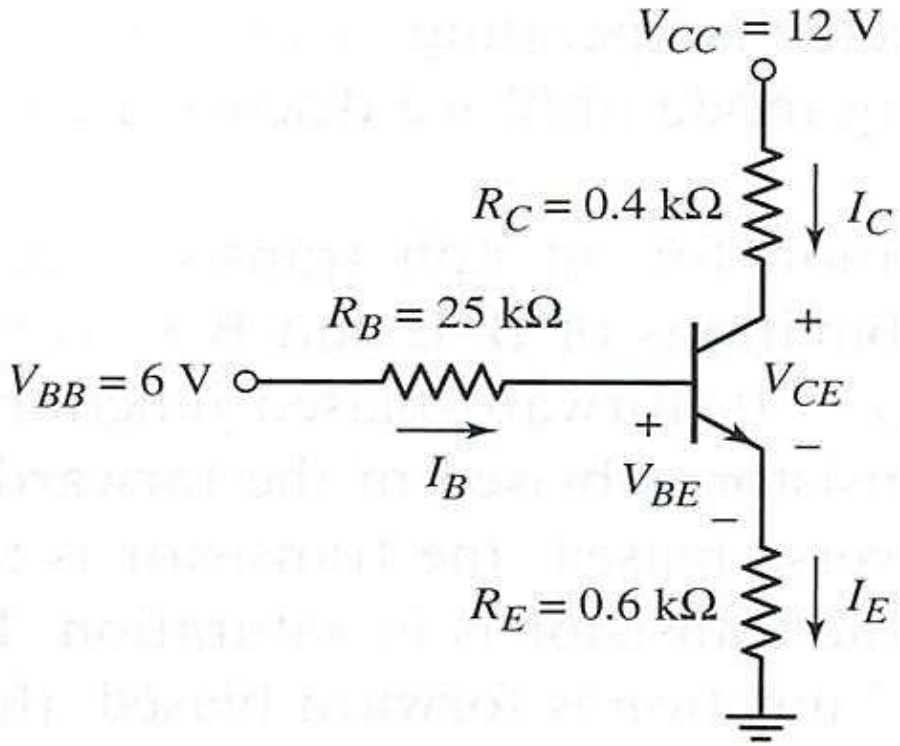
$V_{BE(ON)} = 0.7$ V B = 50

Determine Vo, I_B, I_C

1. Vi = 0.2 V
2. Vi = 3.6 V

Example 4: DC Analysis and Load Line

Calculate the characteristics of a circuit containing an emitter resistor and plot the output load line.
For the circuit, let V_{BE} (on) = 0.7 V and $\beta = 75$.



Output Load Line

Use KVL at B-E loop to find the value of I_B

Solution:

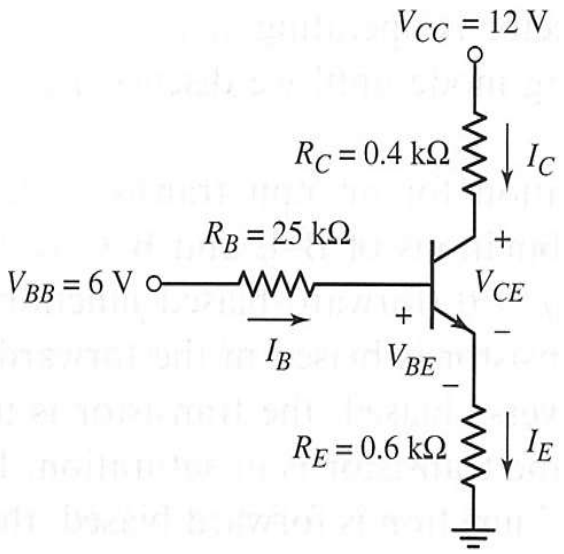
***Q*-Point Values:**

Writing the Kirchhoff's voltage law equation around the B–E loop, we have

$$V_{BB} = I_B R_B + V_{BE}(\text{on}) + I_E R_E$$

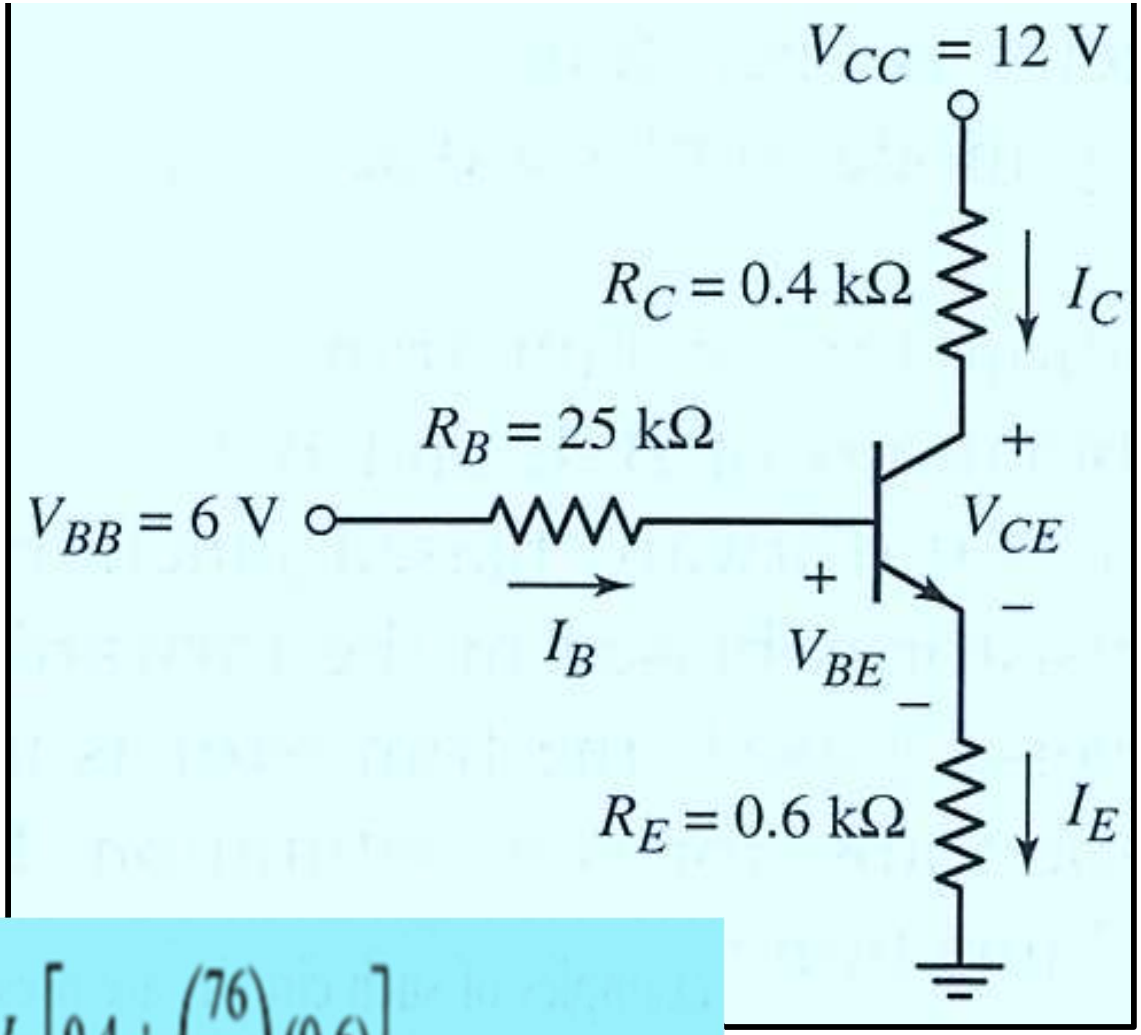
Assuming the transistor is biased in the forward-active mode, we can write $I_E = (1 + \beta)I_B$. We can then solve for the base current:

$$I_B = \frac{V_{BB} - V_{BE}(\text{on})}{R_B + (1 + \beta)R_E} = \frac{6 - 0.7}{25 + (76)(0.6)} \Rightarrow 75.1\,\mu\text{A}$$



Use KVL at C-E loop – to obtain the linear equation

$$Vcc - I_C R_C - V_{CE} - I_E R_E = 0$$



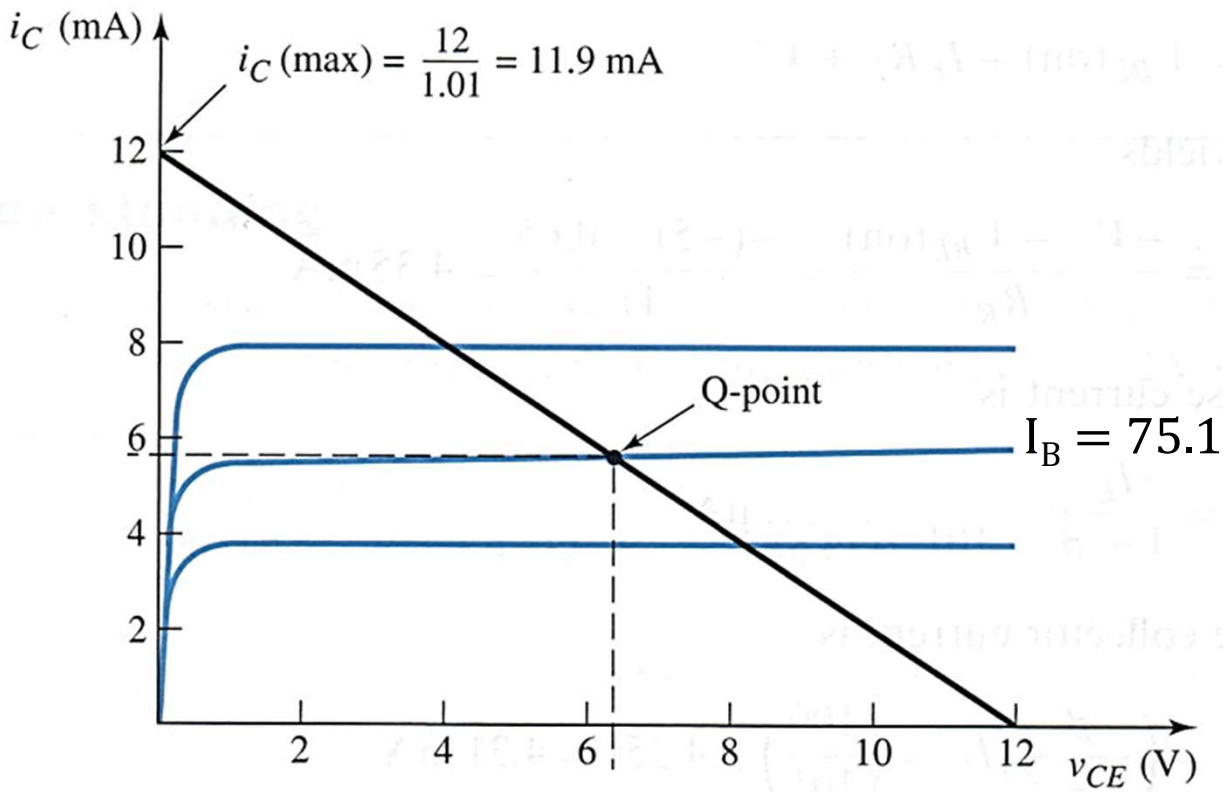
$$V_{CE} = V_{CC} - I_C\left[R_C + \left(\frac{1+\beta}{\beta}\right)R_E\right] = 12 - I_C\left[0.4 + \left(\frac{76}{75}\right)(0.6)\right]$$

or

$$V_{CE} = 12 - I_C(1.01)$$

$V_{CE} = 12 - I_C\,(1.01)$

$I_C = (-V_{CE} + 12) / 1.01$



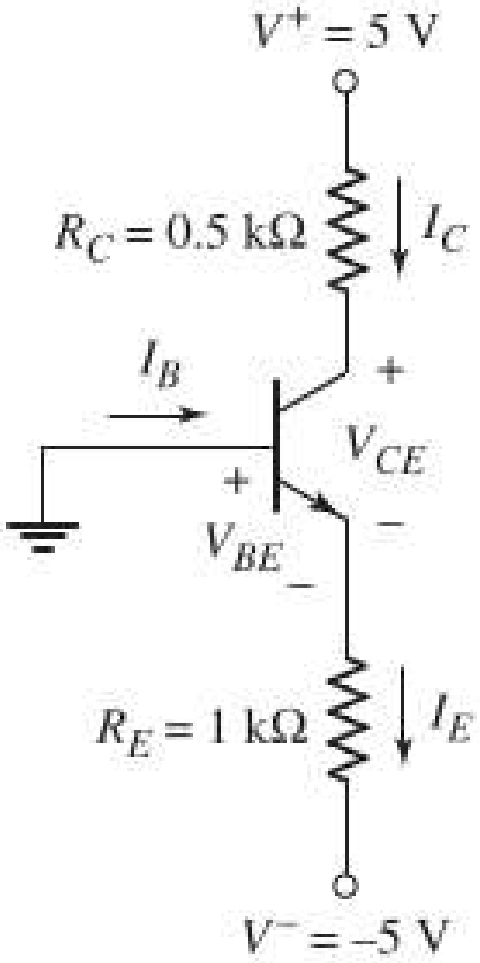
$I_C = \beta I_B = (75)(75.1\,\mu A)$

$\Rightarrow 5.63\,mA$

$V_{CE} = 12 - 5.63\,(1.01)$

$V_{CE} = 6.31\ V$

$V_{BE}(\text{on}) = 0.65$ V and $\beta = 100$.



Writing the Kirchhoff's voltage law equation around the B–E loop,

$$0 = V_{BE}(\text{on}) + I_E R_E + V^-$$

which yields

$$I_E = \frac{-V^- - V_{BE}(\text{on})}{R_E} = \frac{-(-5) - 0.65}{1} = 4.35 \text{ mA}$$

The base current is

$$I_B = \frac{I_E}{1+\beta} = \frac{4.35}{101} \Rightarrow 43.1\ \mu\text{A}$$

and the collector current is

$$I_C = \left(\frac{\beta}{1+\beta}\right) I_E = \left(\frac{100}{101}\right) \cdot (4.35) = 4.31 \text{ mA}$$

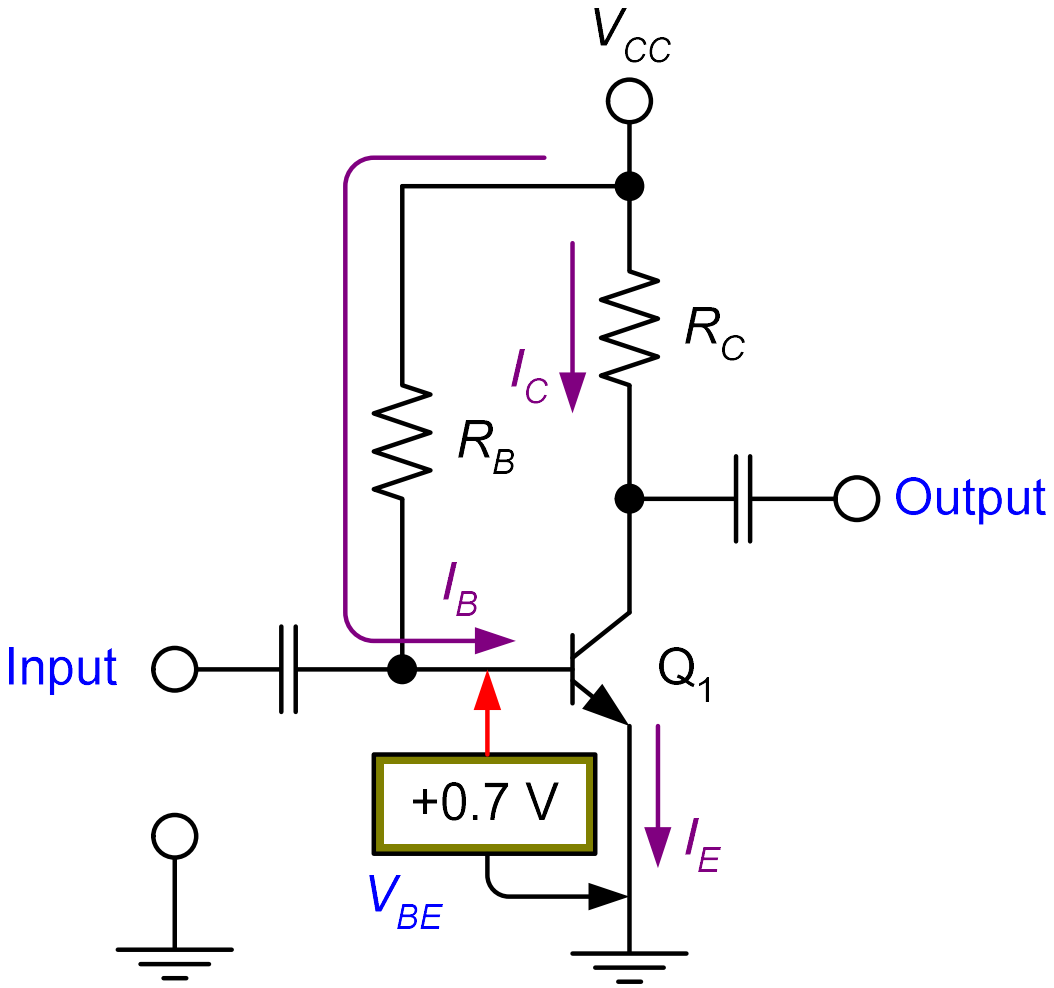
Referring to Figure 5.32(b), the C–E voltage is

$$V_{CE} = V^+ - I_C R_C - I_E R_E - V^-$$

or

$$V_{CE} = 5 - (4.31)(0.5) - (4.35)(1) - (-5) = 3.50 \text{ V}$$

Base bias (fixed bias)



$$V_{CC} - I_B R_B - V_{BE} = 0$$

$$I_B = \frac{V_{CC} - V_{BE}}{R_B}$$

$$I_C = \beta I_B$$

$$V_{CC} - I_C R_C - V_{CE} = 0$$

$$V_{CE} = V_{CC} - I_C R_C$$

β = dc current gain = h_{FE}

$I_{CBO}\uparrow \quad I_C\uparrow \quad T\uparrow \quad I_{CBO}\uparrow$

$I_C = \beta I_B + (\beta + 1)\, I_{CBO}$

Fixed Bias circuit is seldom used

Fixed Bias circuit is seldom used

Effect of I_{CBO} on Stability

- I_{CBO} is the reverse leakage current that flows between the collector and base terminals when the emitter is open.

1. This current is primarily due to the thermally generated minority carriers in the collector-base junction.
   - **Temperature Dependence:** I_{CBO} is highly temperature-dependent. It approximately **doubles for every 10°C rise in temperature**.
   - Impact on Collector Current: The total collector current is given by:

     $I_C=\beta I_B+(1+\beta)I_{CBO}$

     In a fixed bias circuit, I_B is relatively constant (determined by V_{CC}, R_B, and V_{BE}). However, as temperature increases, I_{CBO} increases significantly. Because of the (1+β) term, this increase in I_{CBO} is greatly amplified and directly adds to the collector current.

Effect of I_{CBO} on Stability

1. This current is primarily due to the thermally generated minority carriers in the collector-base junction.
   - **Temperature Dependence:**
   - **Impact on Collector Current:**
   - **Thermal Runaway:** The increase in I_C due to rising I_{CBO} leads to increased power dissipation ($P_D=V_{CE}\times I_C$) at the collector-base junction. This increased power dissipation further raises the junction temperature, which in turn causes I_{CBO} to increase even more. This creates a positive feedback loop known as **thermal runaway**. If not checked, thermal runaway can cause the collector current to rise uncontrollably, eventually damaging or destroying the transistor.

Dependence on β (Current Gain)

2. **Dependence on β (Current Gain):** Beyond I_{CBO}, the fixed bias circuit also suffers from direct dependence on β.

- β varies significantly from transistor to transistor (even of the same type) and also changes with temperature.

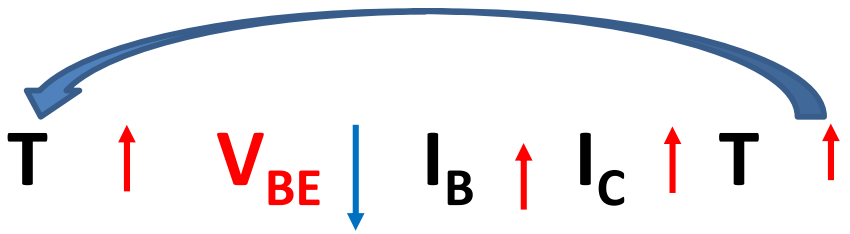
  Since $I_C=\beta I_B$, any variation in β directly shifts the operating point (Q-point) of the transistor, leading to unpredictable performance and potential distortion of the amplified signal.

Sensitivity to V_{BE} (Base-Emitter Voltage)

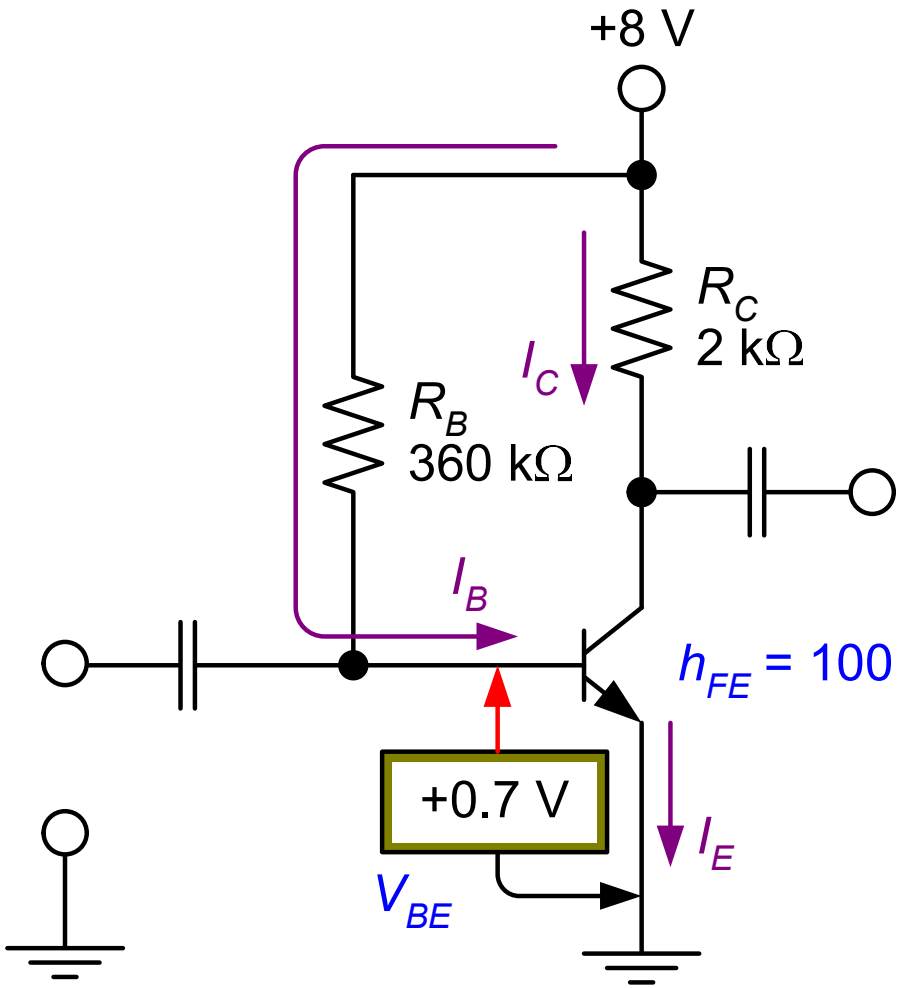
3. **Sensitivity to V_{BE} (Base-Emitter Voltage):**

V_{BE} also varies with temperature (decreases by approximately 2.5mV/°C for silicon transistors). In a fixed bias circuit, a decrease in V_{BE} (due to temperature increase) would cause

$I_B=(V_{CC}- V_{BE})/R_B$ to increase, further contributing to an increase in I_C and worsening thermal instability.



$h_{FE} = \beta$



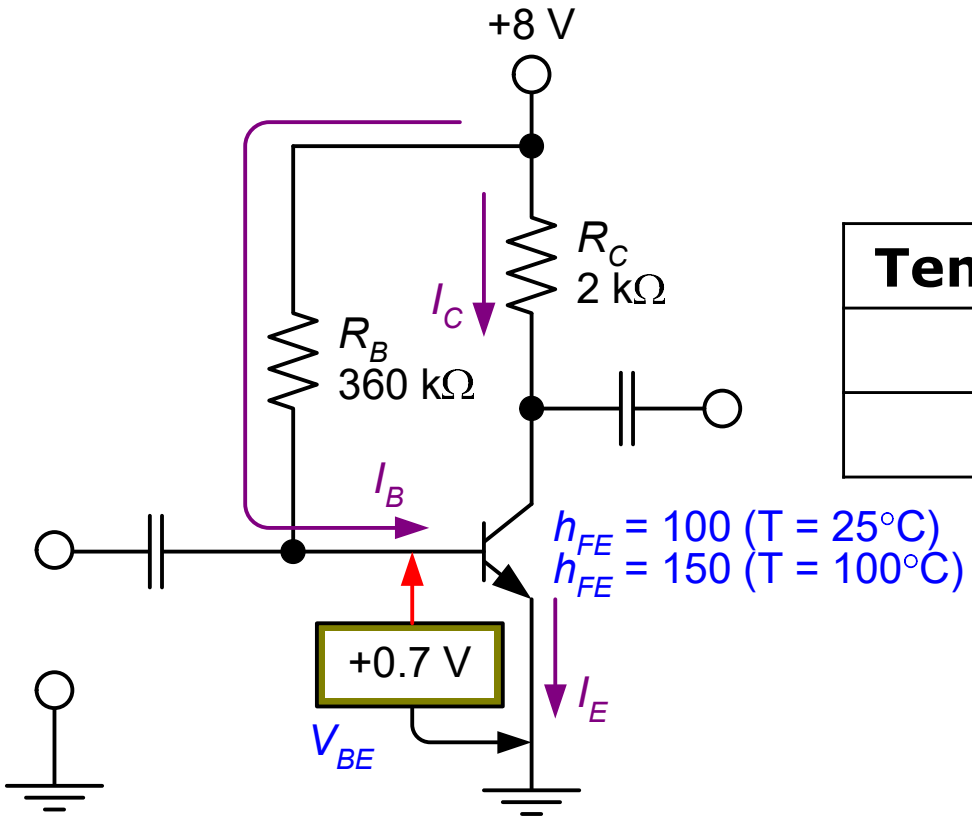
$$I_B = \frac{V_{CC} - 0.7\text{V}}{R_B} = \frac{8\text{V} - 0.7\text{V}}{360\text{k}\Omega}$$
$$= 20.28\mu\text{A}$$

$$I_C = \beta I_B = (100)(20.28\mu\text{A})$$
$$= 2.028\text{mA}$$

$$V_{CE} = V_{CC} - I_C R_C$$
$$= 8\text{V} - (2.028\text{mA})(2\text{k}\Omega)$$
$$= 3.94\text{V}$$

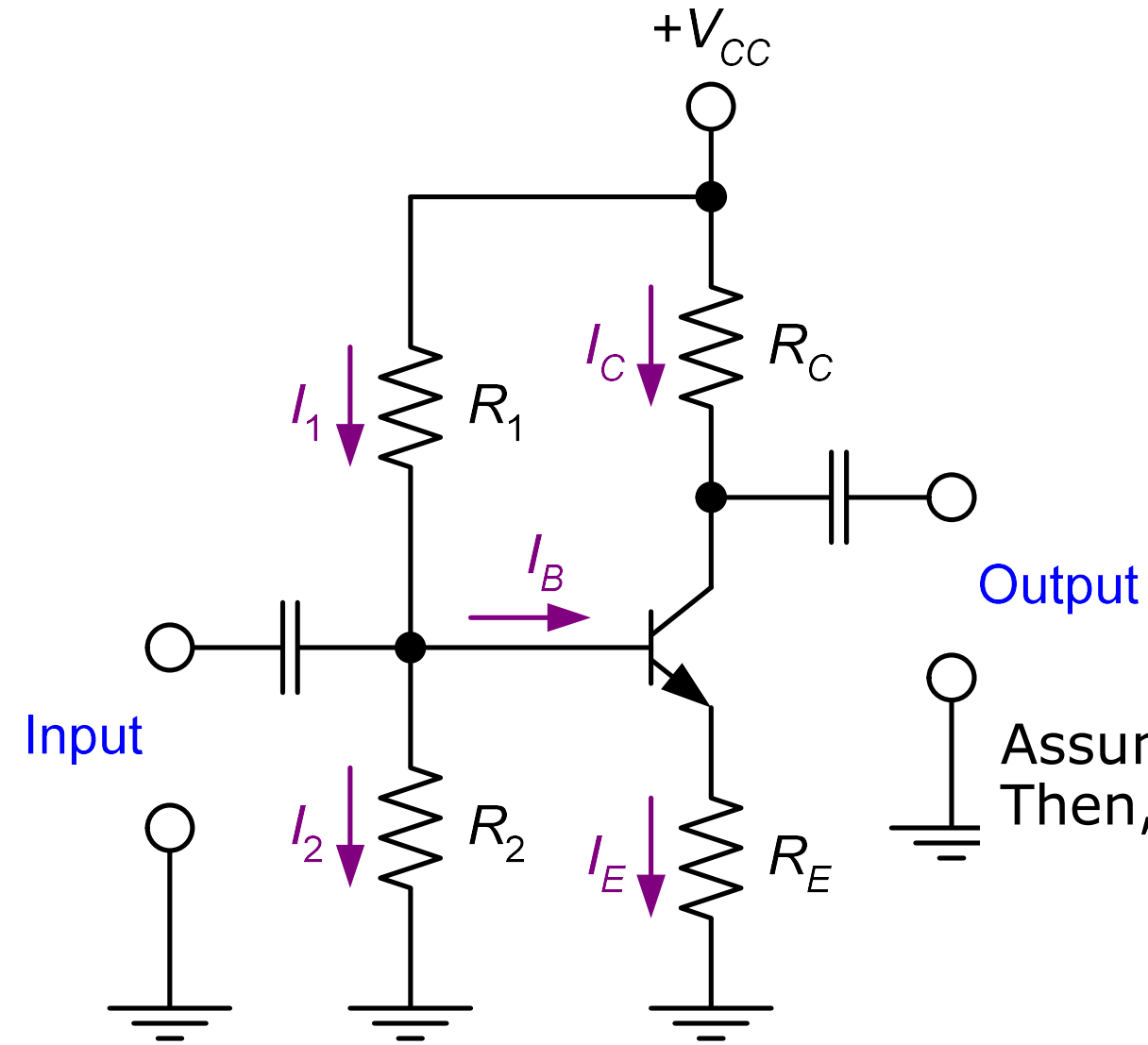
The circuit is midpoint biased.

The transistor in Fig. has values of β = 100 when T = 25 °C and β= 150 when T = 100 °C. Determine the Q-point values of I_C and V_{CE} at both of these temperatures.



Temp(°C)	I_B (μA)	I_C (mA)	V_{CE} (V)
25	20.28	2.028	3.94
100	20.28	3.04	1.92

Voltage Divider Bias



$$V_B = \frac{R_2}{R_1 + R_2} V_{CC}$$

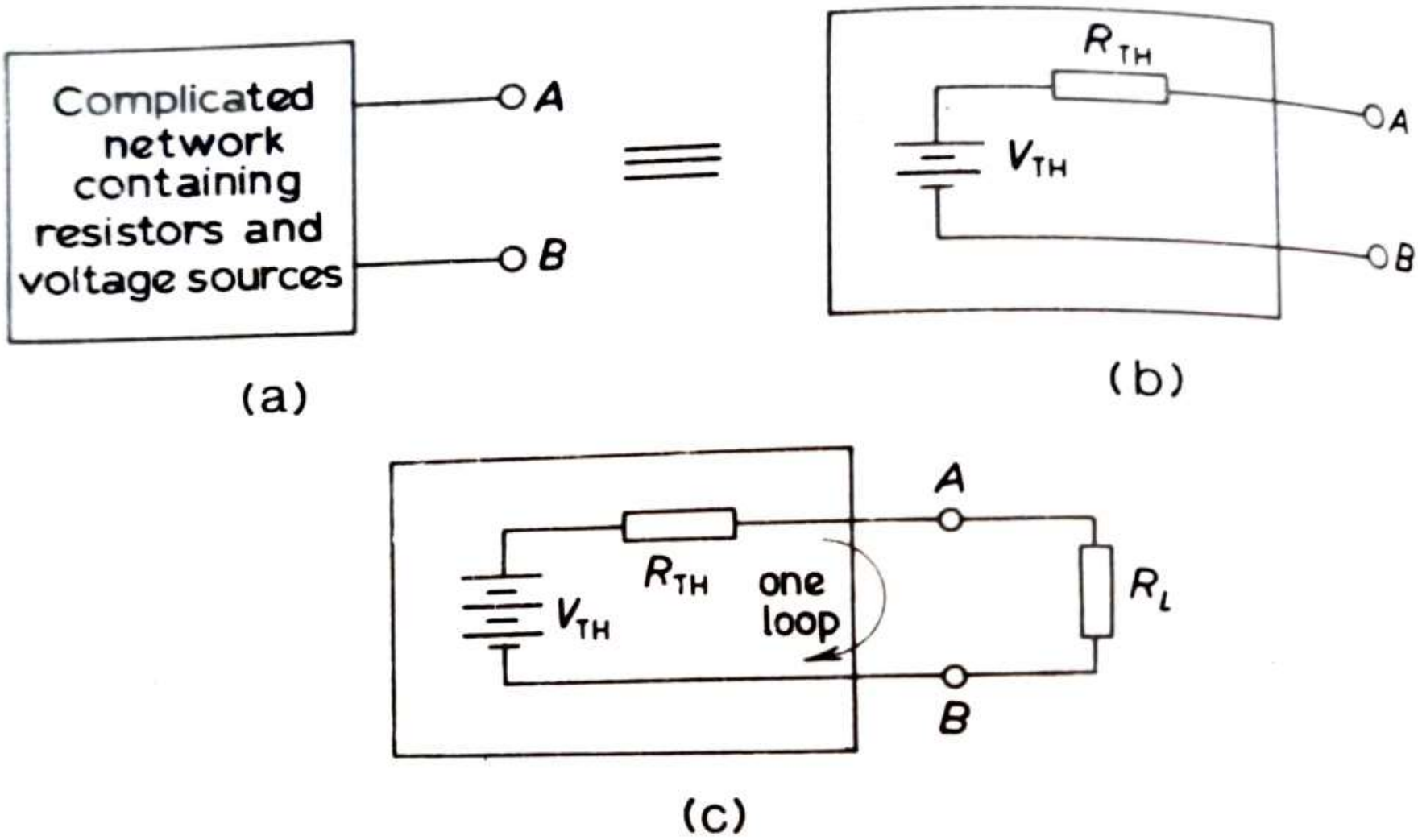
$$V_E = V_B - 0.7\text{V}$$

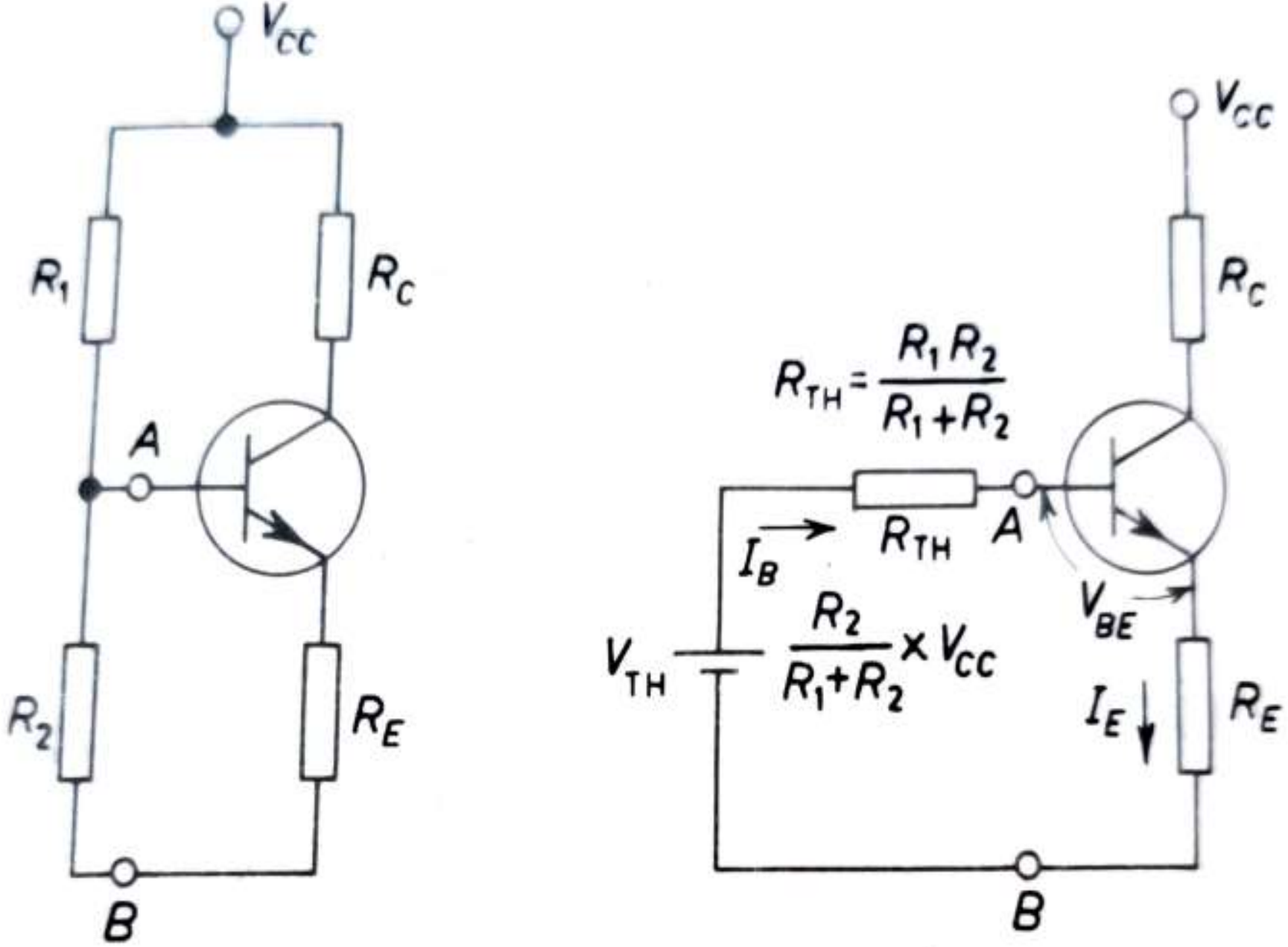
$$I_E = \frac{V_E}{R_E}$$

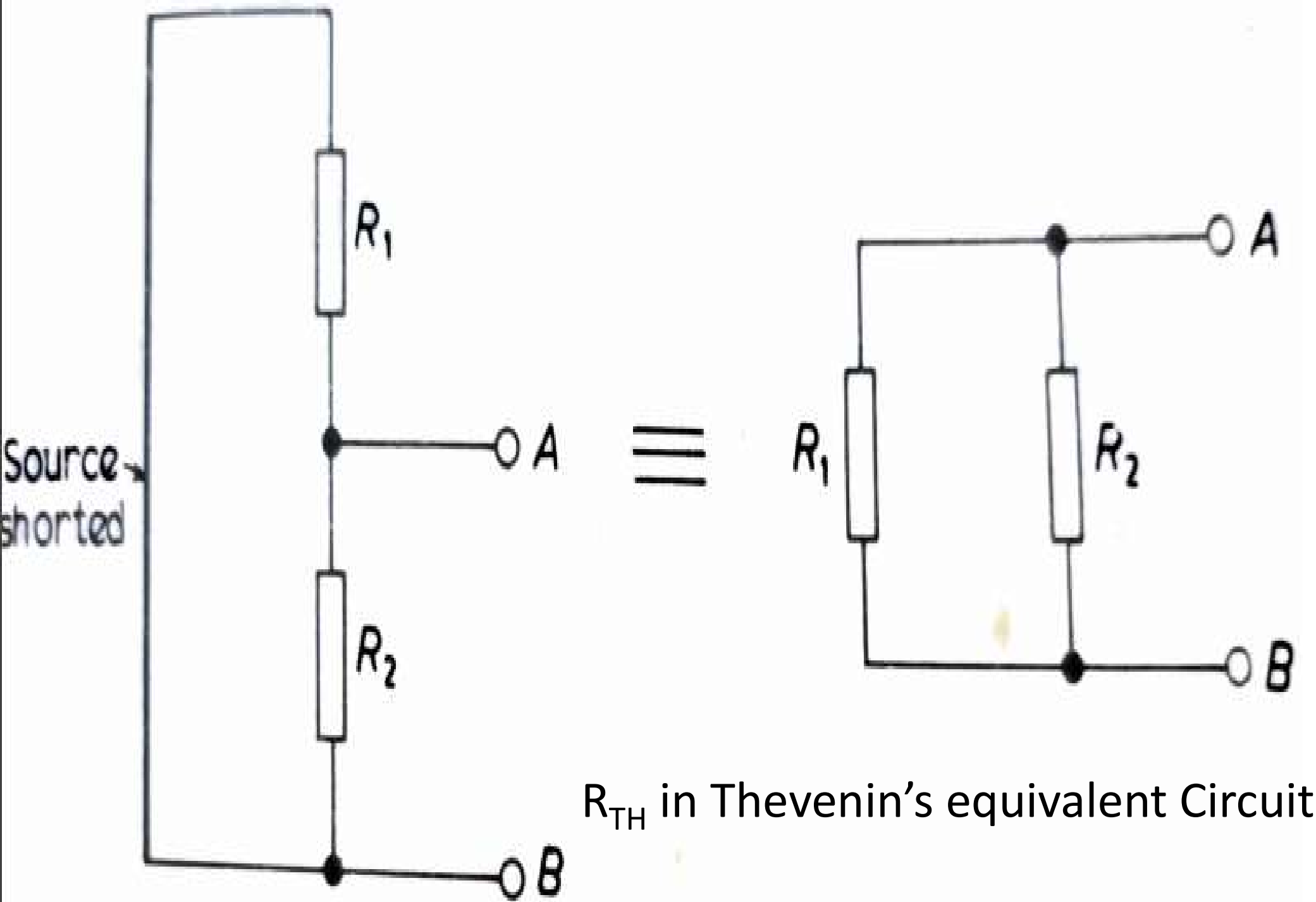
Assume that $I_{CQ} \cong I_E$ (or $\beta >> 1$). Then,

$$V_{CEQ} = V_{CC} - I_{CQ}\left(R_C + R_E\right)$$

Voltage Divider Bias

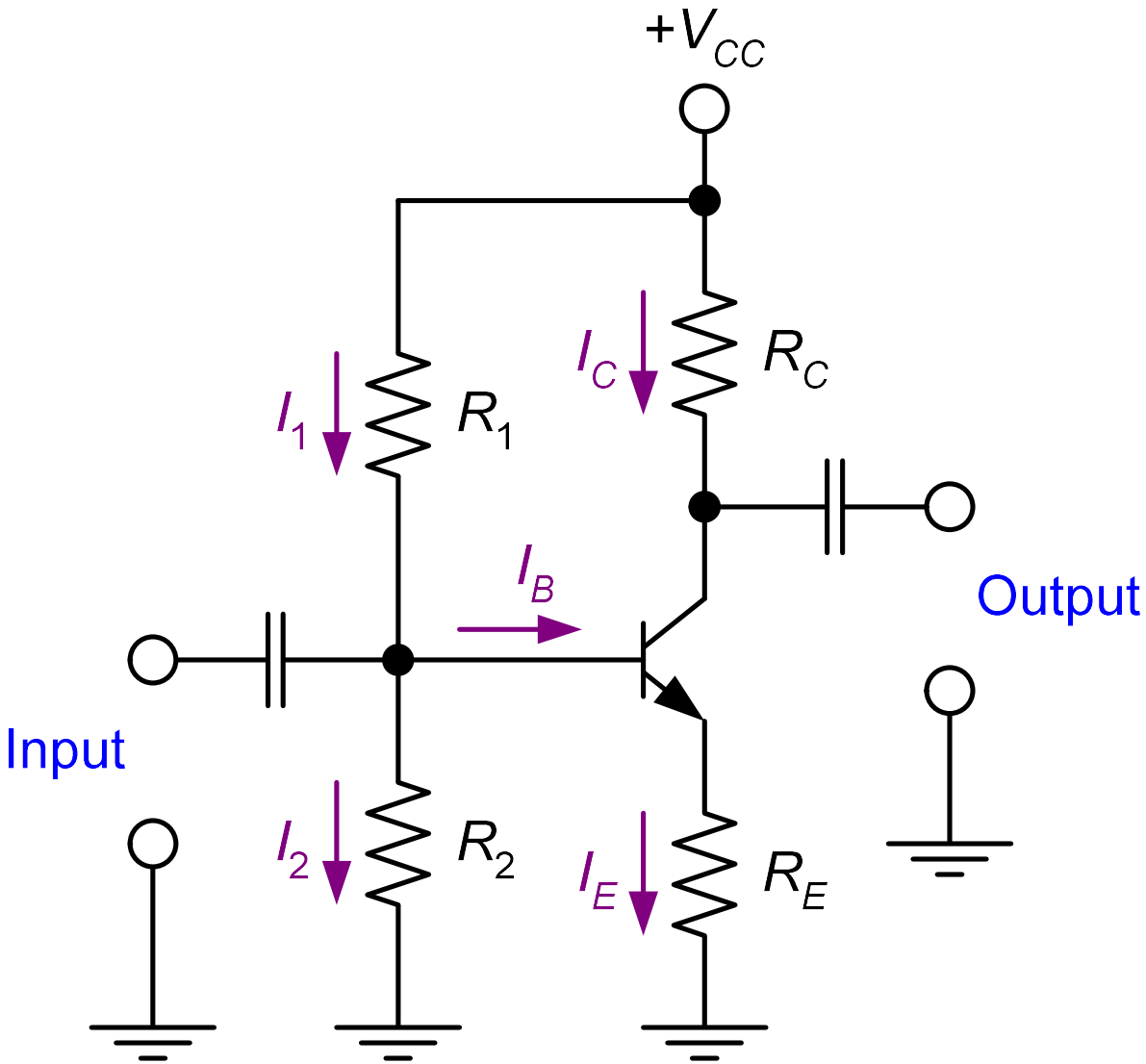



V_{CC}
R_1
R_C
A
R_2
R_E
B
(a)
V_{CC}
R_C
$R_{TH} = \frac{R_1 R_2}{R_1 + R_2}$
R_{TH}
A
I_B
V_{BE}
V_{TH}
$\frac{R_2}{R_1 + R_2} \times V_{CC}$
I_E
R_E
B
(b)

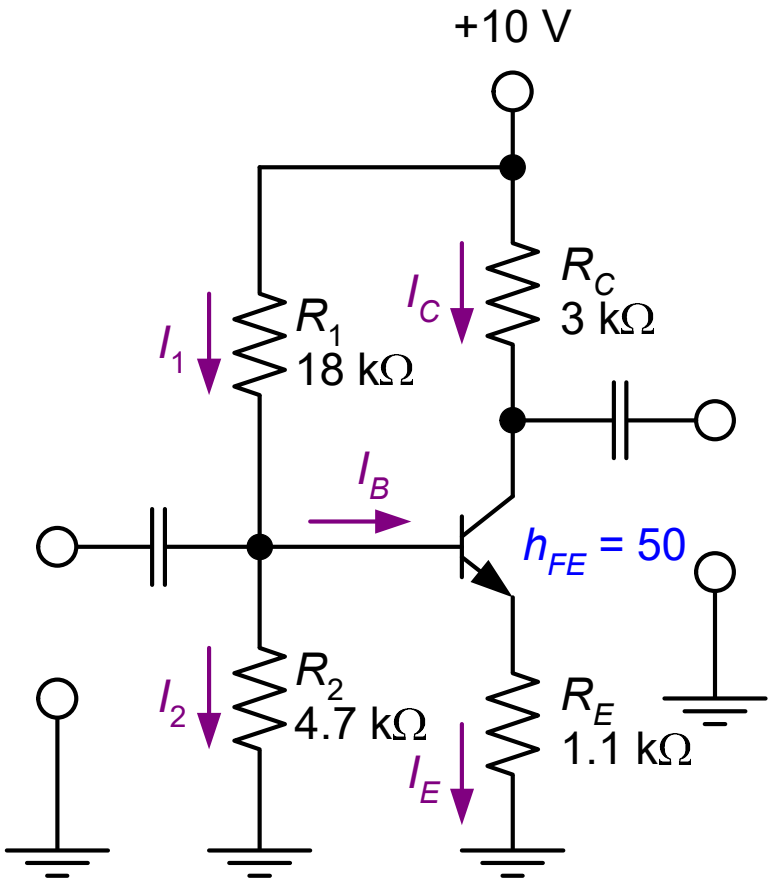


R_{TH} in Thevenin's equivalent Circuit

Voltage Divider Bias



Determine the values of I_{CQ} and V_{CEQ} for the circuit shown in Fig. 7.15.

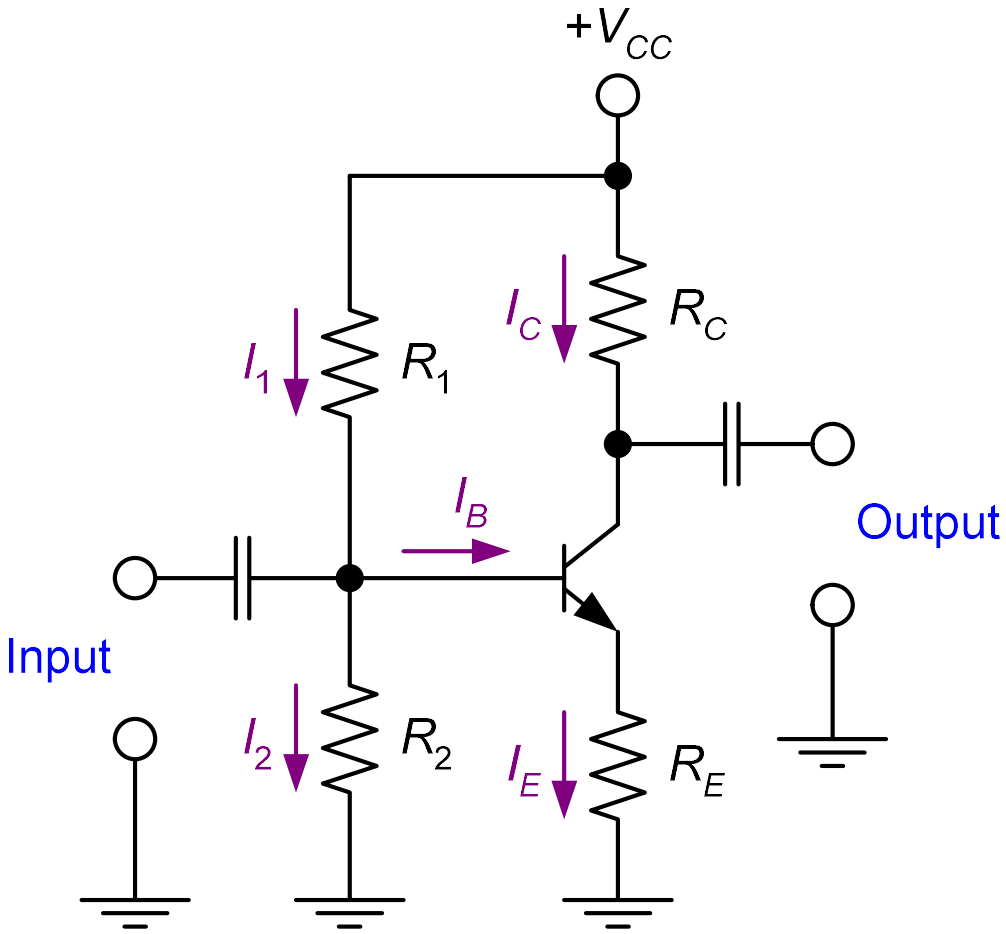


$$V_B = V_{CC}\frac{R_2}{R_1 + R_2}$$

$$= (10\text{V})\frac{4.7\text{k}\Omega}{22.7\text{k}\Omega} = 2.07\text{V}$$

$$V_E = V_B - 0.7\text{V}$$

$$= 2.07\text{V} - 0.7\text{V} = 1.37\text{V}$$



Circuit recognition: The voltage divider in the base circuit.

Advantages: The circuit Q-point values are stable against changes in h_{FE}.

Disadvantages: Requires more components than most other biasing circuits.

Applications: Used primarily to bias linear amplifier.

Other Transistor Biasing Circuits

- Emitter-bias circuits
- Feedback-bias circuits
  - Collector-feedback bias
  - Emitter-feedback bias

Numerical to be solved

- Book by Author Neaman
- TYU 5.1, 5.2, 5.3,5.4,5.5, 5.7,5.8, 5.15,5.16
- Solved example 5.3, 5.4, 5.5, 5.6, 5.7, 5.8, 5.10, 5.15, 5.16, 5.17, 5.18, 5.19
- EX 5.3, 5.4, 5.5, 5.7,5.8, 5.10, 5.15, 5.16, 5.17, 5.18, 5.19